Shaotian Sun

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EDUCATION

University of Michigan, Ann Arbor — Aug. 2024 – Dec. 2026
BS in Honors Mathematics, Minor in CS — *Ann Arbor, MI*
- Cumulative GPA: 3.97/4.00

Shanghai Jiao Tong University — Sept. 2022 – Aug. 2027
BEng in Electrical & Computer Engineering — *Shanghai, China*
- Cumulative GPA: 3.75/4.00 (Top 10%)

RESEARCH EXPERIENCES

High-Dimensional Pairwise Comparison Model — Apr. 2026 - Present
Supervised by Prof. Gongjun Xu — *Dept. of Stats, University of Michigan*
- Investigate statistical estimation and identifiability in high-dimensional pairwise comparison models with low-rank latent interaction structures.
- Developed a penalized likelihood framework to address intrinsic non-identifiability, decomposing parameter perturbations into identifiable and gauge directions and analyzing the corresponding Hessians.
- Established local strong convexity and Hessian invertibility under sparse observation regimes.

Conditional Exponential Directed Last Passage Percolation with Flat Initial Condition — May 2026 - Jul. 2026
Dept. of Math, University of Michigan
Supervised by Prof. Jinho Baik
- Investigate conditional exponential directed last passage percolation with flat initial condition under a one-point upper large-deviation event, focusing on how rare conditioning alters the macroscopic profile.
- Analyze exact multi-point distribution formulas using contour-integral representations and steepest-descent asymptotics.
- Derived conditional near-diagonal fluctuation theorem suggesting Brownian-bridge components coupled with an additional Gaussian mode, and heuristically extended to fluctuations for off-diagonal points.

Computing Tools for Translation-Invariant Total Orders — Aug. 2025 - Jul. 2026
Supervied by Dr. Grant Barkley — *Dept. of Math, University of Michigan*
- Study TITOs (Translation-Invariant Total Orders) using graph-theoretic and computational methods, with a focus on developing systematic tools for exploring and classifying finite restrictions of these order structures.
- Developed graph-based algorithms for constructing and analyzing compatible order relations and implemented them in the Python package `TITO_Explore`, enabling automated computation and visualization of associated combinatorial structures.

PREPRINT

(indicates equal contribution.)*
Yifan Jing*, Emma Yicheng Lou*, **Shaotian Sun***, Yuxuan Wu*. “Computing Tools for Translation-Invariant Total Orders.” [arXiv link]

TALKS & PRESENTATIONS

Conditional Exponential DLPP with Flat Initial Condition [Slides]
- Young Mathematician Conference, Columbus, OH, 2026.
- University of Michigan REU Seminar, Ann Arbor, MI, 2026.

TITOgraphy: When Order Takes Shape [Poster]
- Undergraduate Research Symposium, Ann Arbor, MI, 2026.
- Lab of Geometry Final Poster Session, Ann Arbor, MI, 2025.

Algorithmic Construction of Weak Orders between Translation-Invariant Total Orders (TITOs) [Slides]

- Lab of Geometry Midterm Presentation, Ann Arbor, MI, 2025.

Is Factorization Always Unique? A Journey into Algebraic Number Theory [Slides]

- Directed Reading Program Presentation Session, Ann Arbor, MI, 2025.

CONFERENCES, WORKSHOPS & SEMINARS ATTENDED

2026 University of Michigan REU Seminar, University of Michigan, Ann Arbor, MI

2026 University of Michigan Random Matrix Theory Summer School, University of Michigan, Ann Arbor, MI

UP GRADe, University of Pennsylvania, Philadelphia, PA

2026 Byrne Conference on Stochastic Analysis in Finance and Insurance, University of Michigan, Ann Arbor, MI

SKILLS

Programming: Python (Pytorch), LaTeX, C++, STATA, Matlab, C, Git, Linux

Languages: Chinese (native), English (TOEFL: 110 [R30/L27/S25/W28]), German (beginner)

RELEVANT COURSEWORK

Graduate-Level Math: Probability Theory (A+), Measure Theory and Functional Analysis (A), Complex Analysis (A), Graph Theory (A+), Commutative Algebra (A).

Undergraduate Math: Differentiable Manifolds (A+), Multi-Variable Analysis (A), Single-Variable Analysis (A+), Linear Algebra (A), Honors Abstract Algebra II (A+), Honors Abstract Algebra I (A), Differential Equations (A), Discrete Math (A+).

Stats & Computing: Probability & Statistics (A+), Econometrics (A), Machine Learning (A), Web Database (A-), Theoretical Computer Science (A), Data Structure and Algorithm (A), Signal and System (A).

DIRECTED & INDEPENDENT READING

Representation Theory: A First Course (Harris & Fulton) — Aug. 2025 - Apr. 2026
Mentor: Prof. Griffin Wang — *Independent Reading*

Number Fields (Marcus) — May 2025 - Aug. 2025
Mentor: Ritwick Bhargava — *Directed Reading Program*

TEACHING EXPERIENCES

Course Assistant, MATH 201: Introduction to Mathematical Writing, Winter 2026, *University of Michigan*

Kiluk Tutor, MATH 217: Linear Algebra, Winter 2026, *University of Michigan*

Grader, MATH 525: Probability Theory, Fall 2025, *University of Michigan*

Grader, MATH 217: Linear Algebra, Winter 2025, *University of Michigan*

Teaching Assistant, TC 3000J: Technical Communication, Summer 2024, *Shanghai Jiao Tong University*

Teaching Assistant, TC 3000J: Technical Communication, Spring 2024, *Shanghai Jiao Tong University*

Teaching Assistant, ENGL 1000J: Academic Writing I, Fall 2023, *Shanghai Jiao Tong University*

LEADERSHIP & OUTREACH

Student Advisor, SJTU Global College Advising Center, 2024–2026, *Shanghai Jiao Tong University*

Community Assistant, U(M) Math Corps, 2025, *University of Michigan*

Interview Group Leader, SJTU Global College Interview Group, 2022–2024, *Shanghai Jiao Tong University*

HONORS & AWARDS

Jack McLaughlin Award in Algebra — 2026
James B. Angell Scholar — 2026
M.S. Keeler Dept. of Math Merit Scholarship — 2025 - 2026

University Honors 2024 - 2026
National Undergraduate Scholarship, China 2023 - 2024